EF301 - Industrial Economics

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"Discuss how (i) technological change and (ii) changes in consumer behaviour may impact upon the global automobile industry."

Introduction

In 2022, the world automobile market was worth 2.95 trillion US dollars (Carlier, 2023). The global automobile industry is constantly growing and changing due to various factors. Two of the most important factors to consider are the changes in technology and consumer behaviour. To gain an advantage over competitors, automotive manufacturers must adapt and change their processes and brand identity on a constant basis. This fight to become a market leader is evident with the vast number of individual companies competing within the same market.

The rapid advancement of technology has had an immense impact on the automotive industry. The introduction of hybrid vehicles, electric vehicles (EVs), autonomous technologies, smart features and technology-aided manufacturing have all led to the complex competition between car brands. These car brands continuously release new models with new technology and cheaper production costs, in order to be better than their opponent.

In recent years, consumers have been purchasing with more of an ecological and environmental mindset. This has led to the increase in demand for EVs and cars that release less emissions. Stricter environmental laws by governments have also led to this growth in the EV market, and companies that do not oblige by these laws are penalised. Volkswagen was involved in a NOX emissions scandal where they were fined heavily and lost around 20% of its company value after this scandal was published (Strittmatter & Lechner, 2020).

Changes in technology

The largest advancement in technology within the automotive industry in recent years, is most definitely that of EVs. Many governments' encouragement of purchasing EVs, along with the growing amount of environmentally conscious people, have been catalysts of the growth of the overall sales of EVs. By 2030, it is even estimated that they will account for over 25% of all car sales in the world (Carlier, 2023). With global warming becoming a much greater concern, a global paradigm shift must be made in the way of using renewable energy for everything and stopping the reliance on fossil fuels. This is where EVs prevail over standard internal combustion engines.

The safety of EVs and the batteries used in EVs concern many consumers due to many reasons. Many consumers aren't enticed to buy EVs due to the belief that the long-term sustainability and quality is not yet known. Studies have been conducted regarding EV safety and they have yielded positive results, as these cars have more room for crumple zones, to ensure that the passengers are as safe as possible. There are also concerns over the battery in accidents as it could cause fires or explosions. The studies conducted still led to optimal safety of passengers as the battery packs remained undamaged (Victor Chombo et al. 2021).

The long charging times, expensive repairs to batteries, and short-range batteries have clouded the global EV market, as they are expensive issues to fix. Wireless and fast charging have both been in the works for some

time now and are slowly being integrated into new EV models (Sagar et al. 2023). The battery packs are also becoming smaller, increasingly durable and with a higher power capacity (Liu et al. 2022).

Autonomous vehicles are also relatively new and present a drastic shift in consumer demand. Tesla is one of the front runners in the category of autonomous driving as all of their cars sold after 2014 have come with in-built auto pilot. These autonomous technologies drive vehicles by using artificial intelligence algorithms that are paired with various sensors. The Internet of Things has been developed in the autonomous automobile industry to allow vehicles to communicate between one another and the environment around it, this is necessary to ensure that the vehicles can be fully autonomous (Gopinath et al. 2023). Although this is innovation at its finest, there has been many safety concerns with autonomous vehicles.

One such case occurred in 2019, when the autopilot feature was used in a Tesla Model 3 when it caused a fatal accident. A lawsuit blaming this technology was carried out in 2023 and concluded that in no way was the technology at fault. Tesla's software always requires drivers to be prepared to regain control of the car at any moment, the driver in this case did not do so (Ewing, 2023). It is clear that autonomy in vehicles has not been perfected yet, and any issues that arise in relation to it will taint its appearance to the public.

With the prevalence of transport services, such as Uber and Bolt, it was only a matter of time before these companies eliminate their biggest costs, the drivers. This elimination would drastically increase their profit margins if implemented correctly. Throughout the USA we have already seen the introduction of these fully self-driving vehicles, but they have led to some issues and have come under scrutiny. The inability of these vehicles to respond to irregular on-road scenarios, such as when they encounter emergency service vehicles is a worrying idea. In 2023, in San Francisco, an ambulance was blocked in by 2 driverless vehicles. Luckily, this did not cause the patient to lose his life, but it did slow down the emergency services by 7 minutes. Driverless vehicles like these aren't able to be controlled from elsewhere and must be towed away. This was just one of 600+ driverless vehicle reports that were given in a 12-month period in San Francisco alone (Lu, 2023).

Large scale implementation of unready systems unnecessarily takes up hours of emergency services' hours, and clearly these problems should be ironed out prior to release. The early issues of this driverless technology may have a negative lasting effect with individuals unable to trust such systems due to prior experience. Therefore, with improper preparation it must be believed that this early stage of autonomous vehicles must be taken with more precaution, as the effects of either a positive or negative next few years in this area will have an immense impact on the future global automobile industry.

Another key aspect of technological change in the global automobile industry is the manufacturing processes that companies go through to make these automobiles. Companies regularly adapt to new technologies in order to benefit the most and increase their overall efficiency. A significant growth in the use of AI and robotics in automobile manufacturing has been seen through the last decade, as it yields immense benefits. The use of robotics in the manufacturing of vehicles allows companies to maintain a non-stop customisable operation that is fully under their control, increasing overall efficiency greatly. It also takes away many safety concerns, as machinery can undertake unsafe actions without causing any injuries. Although initial costs are seen as high, the long-term use of robotics reduces staff labour costs and therefore greatly reduces overall costs. Minimal human staff can be seen as a negative in all businesses, it can be argued that replacing humans with robots is unethical as jobs are lost. Robotics can also be expensive to repair and maintain, and new higher-paid workers must be employed that are qualified in a technological field in order to maintain the technology (Prithvi, 2023).

Although there are numerous setbacks when looking at the large-scale implementation of technology in car manufacturing, it is hard to argue against the idea that the benefits heavily outweigh the setbacks. This robotisation in manufacturing will lead companies into a more efficient future and cause them to employ a smaller workforce that has more of an expertise in technology rather than manual labour. This shift in employee requirements will reduce the availability of jobs to unqualified personnel but increase the demand in individuals that are qualified in various fields of technology.

Changes in consumer behaviour

There are numerous ongoing changes in consumer behaviour have a great impact on the worldwide automobile industry. These changes mainly encompass the areas of if a vehicle is necessary to purchase, growth in environmental consciousness, and the growth of online marketing and sales.

The demand for hybrid and EVs by consumers has increasingly grown over recent years, and as the technology is furthermore improved these vehicle types will ultimately takeover the global automobile market. The growth of these sustainable and environmentally friendly alternatives has grown due the increased improvements, constant reduction in prices, increased environmental awareness and the availability of government grants that encourage the purchasing of such vehicles (Muratori et al. 2021). Although much government funding in Ireland has been reduced or even removed for the purchasing of EVs, grants for EVs and EV home chargers still exist (citizensinformation.ie, 2024).

Urbanisation and a higher population density in cities has caused the issue of smog on a global scale, especially in China. China have increased their encouragement of the purchasing of hybrids and EVs, and there was a direct correlation found between less engine powered vehicles and the intensity of smog. Smog is an issue that has worried consumers immensely, as it is carcinogenic, and these cleaner vehicles offer consumers a way to increase air purity and reach for carbon-neutrality (Su et al. 2021).

Another issue that derives from urbanisation is that of traffic congestion and the increased need of efficiency, speed, and convenience for consumers in the automobile industry. In urbanised areas heavy traffic is very common and this leads consumers that live in these areas to consider fuel efficiency when making vehicle purchases. This, along with the need for compactness and convenience to be easily mobile in large cities is a big driver for change in the automotive industry (Deloitte, 2009). With smaller vehicles, that are more fuel efficient there is a saving in costs and time. This change in consumer needs makes companies in the automobile industry focus on providing safe, fuel efficient compact cars that are perfectly suited for larger cities.

Another way to save time and money is, of course, not own a vehicle at all. Companies such as Uber and Zipcar allow people to travel from point A to point B in an inexpensive and convenient manner. Transportation companies are broadening their market by no longer relying solely on car sales, and expanding the after-car-sales market, where vehicles or transport in general can be affordably used as a service whenever needed (Alvarado-Sieng and Huerbsch, 2018). Areas with high population density may be difficult to navigate quickly, parking may be expensive, and it just may be unrealistic to own vehicles in such environments, when there are such convenient alternatives.

Consumers are gradually leaning towards an all-online shopping experience, now simple day-to-day tasks such as grocery shopping can easily be done online. Companies such as Tesla, BMW and Porsche have took this into consideration, and now on their websites you can choose from a range of models and fully customise your vehicle to your liking without having to leave your home. Platforms like company websites and social media are the way to market products to the new generation of customers, as they prefer a quick and easy experience rather than one that requires interactions and energy (Aryani et al. 2021). This change in consumer behaviour stems from the Covid 19 pandemic, where online shopping was advertised more, people stayed at home and worked from home more, and most companies became accustom with working with an online platform. This change allows companies to look to the future and possibly eliminate the need for car-sales people, dealerships and look to moving to a more online based experience where costs can be cut in all aspects of the business.

Conclusion

In conclusion, the global automobile industry is undergoing constant pressures from changes in technology and consumer behaviour.

The development of environmentally friendly vehicles, autonomous driving and robotics in car manufacturing have led the global automobile industry to a new era. At the moment the world is going through a shift into becoming heavily reliant on technology in all aspects of daily life, especially in transportation. While the

technologies involved have got a long way to go before becoming fully perfected and made affordable to all, the substantial growth in the EV market cannot be understated. Many concerns are carried, and these concerns will remain until autonomy and EVs become fully flawless.

Similarly, consumer behaviour has undergone immense change in environmental awareness, urbanisation, and digitalisation. Environmental consciousness seems to be an outstanding leader in what causes change. Consumers' growing need to reach carbon neutrality has caused an immense shift in the automobile market that will only grow substantially over the coming years. The technologies used in obtaining cleaner vehicles have not reached their peak yet and are still relatively expensive compared to their Diesel and Petrol counterparts, which does not make owning an EV possible for everybody at the moment.

Urbanisation has also caused what seems to be a shift away from car ownership due to the lack of convenience, affordability, and sustainability within the current car marker. Consumers are now more enticed by ideas such as carpooling and getting an Uber to work, rather than owning and maintaining their own vehicle. This sway away from car ownership was boosted by companies like Uber that are digitalising the transportation industry, and this may have led to automobile companies undertaking initiatives to greater digitalise their own brands.

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